

Facebook Reels Algorithm and Strategy

<https://gemini.google.com/app/2629d30dc85d14fb>

User prompt: how facebook algo work, how they suggest my reels too audience, what are the areas i should have to care when i am creating reels that could show to vast audience

Response: Facebook's algorithm uses a personalized, AI-driven recommendation process (sometimes called Meta's "Discovery Engine"). Unlike your main feed, which prioritizes posts from friends and followed Pages, **Reels focus on un-connected discovery**—meaning Facebook actively pushes your videos to people who do *not* follow you yet.

How the Facebook Reels Algorithm Works

When you upload a Reel, Facebook scores it against millions of others using a four-step process:

1. **Testing Phase (Early Signals):** Your Reel is shown to a small test group (a mix of active followers and non-followers who like your niche).
2. **Predictive AI Scoring:** Meta evaluates initial signals—such as retention rate, completion rate, shares, and comments—to predict how a wider audience will react.
3. **Interest Matching:** The AI scans video audio, visual elements, captions, and text-on-screen to categorize the topic and serve it to users with matching viewing habits.
4. **Mass Distribution:** If your test group shows strong retention and positive signals, the algorithm scales up distribution exponentially to a broader audience.

Key Ranking Signals (Weighted by Importance)

- **Watch Time & Completion Rate:** The single most critical metric. Did people watch past the first 3 seconds? Did they watch it to the end or loop it?
- **Shares & Send Rates:** When a user shares a Reel via Direct Message (DM) or to their own Story, Meta treats it as a high-intent endorsement, boosting distribution.
- **Meaningful Engagement:** High-value comments (longer discussions) and saves outweigh simple likes.
- **Direct User Surveys:** Facebook periodically prompts users with a pop-up ("Was this Reel worth your time?"). Positive sentiment scores heavily influence organic reach.

Critical Areas to Focus On for Maximum Reach

1. The First 3 Seconds (The Hook)

- **Visual Action:** Start in the middle of the action or movement. Avoid static setups, slow intros, or saying "Hey guys."
- **On-Screen Text & Audio:** Place clear text overlays in the upper middle area to catch viewers who watch on mute.

2. Technical Quality & Format Specs

- **Strict Vertical Ratio:** Shoot and edit strictly in **9:16 (1080 x 1920 px)**. Horizontal videos with black bars are heavily suppressed.
- **High Resolution:** Upload crisp 1080p, 30/60fps MP4 video. Avoid blurriness, as low bitrate triggers negative quality penalties.
- **No Watermarks:** Meta actively demotes videos containing watermarks or logos from competitor platforms (like TikTok or CapCut logos).

3. Strict Originality (Meta's Top Priority)

- **Avoid Reposting Generic Clips:** Re-uploading meme clips, TV shows, or unedited stock footage leads to algorithmic reach restrictions.
- **Transformative Edits:** If using third-party audio or video clips, add substantial value (e.g., your voiceover, reaction commentary, or unique editing).

4. Engineering "Loopability" and Shares

- **Pacing & Length:** Keep Reels under 30–60 seconds unless the story warrants more. Faster-paced edits keep drop-off low.
- **Seamless Loops:** Design the end of your video to flow smoothly back into the beginning. Automatic rewatches signal high retention to the algorithm.
- **Relatability Factor:** Ask yourself, *"Why would someone send this to a friend?"* Content that offers a quick tip, a relatable joke, or an surprising insight generates high share rates.

5. Optimizing for "Social Search"

- **Keyword-Rich Captions:** Treat your caption like SEO. Use natural language and niche-specific keywords so Meta's visual-text models can index your topic accurately.

- **Relevant Trending Audio:** Use trending sounds directly from the Facebook Audio Library to leverage existing traffic.

Sources

1. [How the Facebook Algorithm Works in 2026 for Organic Reach](#)
2. [How Does the Facebook Algorithm Work in 2026? - Conbersa](#)
3. [What Are Facebook Reels? Format, Algorithm, and Strategy Guide](#)
4. [Facebook's 2026 Rules for Reach & Relevance](#)
5. [Facebook algorithm explained: 2026 insights - SocialBee](#)
6. [Facebook Reels Algorithm Explained - Dash Social](#)

User prompt: Give me 5 proven hook templates for Facebook Reels designed to capture attention in the first 3 seconds.

Response: To win the first 3 seconds on Facebook Reels, a hook must pair a **verbal/textual trigger** with a **visual movement**.

1. The "Negative Warning" Hook

People are far more motivated to avoid a mistake than to receive positive advice.

- **Text/Spoken Template:** *"Stop doing [Common Industry Practice] if you want [Desired Result]!"*
- **Example:** *"Stop posting standard photos if you want to grow your local bakery business in 2026."*
- **Visual pairing:** A bold gesture like leaning in quickly toward the camera or a fast "X" motion across the screen.

2. The Curiosity Gap

This template creates an immediate knowledge gap that forces the viewer to stay to get the answer.

- **Text/Spoken Template:** *"Why is nobody talking about this [Tool / Method / Trick] for [Specific Goal]?"*
- **Example:** *"Why is nobody talking about this hidden audio setting that doubles Reel reach?"*
- **Visual pairing:** Pointing directly at a text box, or opening a laptop/phone with a shocked facial expression.

3. The "Us vs. Them" Pattern Interrupt

By directly challenging standard wisdom, you instantly interrupt passive scrolling.

- **Text/Spoken Template:** *"Most people think [Common Belief], but here is what actually works..."*
- **Example:** *"Most people think you need 10,000 followers to land sponsorships, but here's what actually works..."*
- **Visual pairing:** Holding an item related to your niche and abruptly setting it aside or doing a fast cuts transition.

4. The Transformation Tease

This taps into immediate visual proof, making the audience want the same result.

- **Text/Spoken Template:** *"How I went from [Negative Starting Point] to [Positive Result] in [Short Timeframe] using ONE simple change."*
- **Example:** *"How I went from zero views to 50k weekly reach in 14 days using ONE editing rule."*
- **Visual pairing:** Show the high-end result or metric screenshot in the background during the first 1.5 seconds, then cut directly to your face.

5. The Relatable "Exposing Secrets" Hook

Positions you as an insider giving away gatekept information.

- **Text/Spoken Template:** *"The [Industry / Topic] secret that [Target Group] doesn't want you to know..."*
- **Example:** *"The social media secret that big marketing agencies don't want you to know..."*
- **Visual pairing:** Speaking in a lowered, "whisper" tone close to the microphone or camera to build immediate intimacy.

Best Practices for All 5 Hooks

- **Front-load the power words:** Place action words (e.g., *Stop*, *Secret*, *Proven*, *Avoid*) within the first 1.5 seconds.
- **Keep text in the safe zone:** Ensure text overlays sit in the upper-middle region so Facebook UI buttons (like, comment, share) don't cover the text.
- **Match audio to motion:** Pair the spoken hook with a sharp audio sound effect (a pop, swoosh, or chime) to grab muted and unmuted viewers alike.

Sources

1. [Facebook Reels Hook Formulas That Drive 3-Second Holds](#)
 2. [280 Short Video Hook Templates | PDF - Scribd](#)
 3. [100 Best Viral Video Hooks 2024 - Online Path](#)
-

User prompt: Write 3 complete short-form Reel scripts (under 30 seconds each) built around these hook templates.

Response: Here are 3 complete, 30-second Facebook Reel scripts engineered for high retention, screen placement, and strong hook-to-call-to-action (CTA) flow.

Script 1: The "Negative Warning" Hook

Target Goal: High Shares & Engagement (Educational / Tips)

Estimated Duration: 20–25 seconds

- **[0:00 - 0:03] THE HOOK**
 - **Visual:** Close-up of your face, leaning into the camera quickly with hand raised in a "stop" motion.
 - **On-Screen Text:** *STOP DOING THIS IF YOU WANT REEL VIEWS!*
 - **Audio/Voiceover:** *"Stop posting horizontal videos to Facebook Reels if you actually want to grow your reach!"*
- **[0:03 - 0:15] THE VALUE / PROBLEM**
 - **Visual:** Fast cut to a screen recording showing a 16:9 horizontal video with huge black bars at the top and bottom, highlighted with red "X" icons.
 - **Audio/Voiceover:** *"When you upload horizontal clips, Facebook automatically shrinks your video frame, cutting your visual real estate by more than 50%."*
- **[0:15 - 0:22] THE SOLUTION**
 - **Visual:** Cut back to you holding a phone vertically, filling the whole screen (9:16 ratio) with a green checkmark graphic.
 - **Audio/Voiceover:** *"Always crop or film strictly in 9:16 vertical ratio. Full-screen content gets 3x higher watch retention on the algorithm."*
- **[0:22 - 0:28] THE CALL TO ACTION (CTA)**
 - **Visual:** On-screen text popup pointing down.
 - **Audio/Voiceover:** *"Send this to a creator who needs to fix their formatting, and follow for daily growth tips!"*

Script 2: The Curiosity Gap

Target Goal: High Retention & Comments

Estimated Duration: 20–25 seconds

- **[0:00 - 0:03] THE HOOK**
 - **Visual:** Pointing up at bold text while giving a genuine "shocked" or confused expression.
 - **On-Screen Text:** *Why is NOBODY talking about this audio trick?*
 - **Audio/Voiceover:** *"Why is nobody talking about this hidden audio setting that doubles your Reel reach?"*
- **[0:03 - 0:12] THE STEP-BY-STEP VALUE**
 - **Visual:** Rapid screen recording of your phone screen zooming into the Facebook Reels audio selection tool.
 - **Audio/Voiceover:** *"Before you publish, open the Audio tab, click 'Trending Sounds', and pick a track with an arrow pointing up next to it."*
- **[0:12 - 0:20] THE "WHY IT WORKS"**
 - **Visual:** Cut to a quick text overlay over dynamic background footage: *"Meta favors trending audio signals."*
 - **Audio/Voiceover:** *"Meta deliberately pushes videos that use rising sounds to test early audience reactions faster."*
- **[0:20 - 0:26] THE CALL TO ACTION (CTA)**
 - **Visual:** Pointing to the comment section on screen.
 - **Audio/Voiceover:** *"Drop the word 'AUDIO' in the comments and I'll send you the top 5 trending tracks this week!"*

Script 3: The Transformation Tease

Target Goal: High Save Rate & Viral Distribution

Estimated Duration: 25–30 seconds

- **[0:00 - 0:03] THE HOOK**
 - **Visual:** Flash a high-performing analytics graph or screenshot on screen for 1 second, then cut back to you speaking directly to the camera.
 - **On-Screen Text:** *0 to 50k Views in 14 Days (Using 1 Rule)*
 - **Audio/Voiceover:** *"How I went from under 100 views per Reel to over 50k reach in two weeks using ONE editing rule."*
- **[0:03 - 0:14] THE INSIGHT**
 - **Visual:** Side-by-side editing demo showing slow pacing versus fast visual cuts every 2 seconds.
 - **Audio/Voiceover:** *"I stopped talking for more than 3 seconds without changing the visual on screen. B-roll, zooms, and text transitions keep the eye locked in."*
- **[0:14 - 0:22] THE AUDIENCE TAKEAWAY**
 - **Visual:** Fast sequence showing 3 visual switches in 5 seconds (Text overlay, camera zoom, graphic popup).
 - **Audio/Voiceover:** *"If your video stays static for 5 seconds, 80% of viewers scroll away. Motion equals watch time."*
- **[0:22 - 0:28] THE CALL TO ACTION (CTA)**
 - **Visual:** Text overlay: *SAVE THIS REEL FOR YOUR NEXT EDIT*
 - **Audio/Voiceover:** *"Save this video so you remember to apply this rule to your next edit!"*

Key Production Tips for All 3 Scripts

1. **Pacing:** Keep spoken speed snappy (130–150 words per minute).
2. **On-Screen Text Placement:** Keep text in the center third of the screen so it isn't covered by Facebook's like/comment icons on the right or caption on the bottom.
3. **Sound Effects:** Add subtle "swoosh" or "pop" audio triggers on every text overlay to hold user focus.

User prompt: what i am doing is gather data around the topic and give the details to the notebookLLM to create reel video, and publish it, my topics and this fb page regarding the Science and the technology information sharing. so what are thing i should follow to grow my audience

Response: Using NotebookLM as a research and video-generation engine is a powerful, efficient workflow for Science and Technology content. However, because raw AI outputs can sometimes feel synthetic or dense, success on Facebook comes down to **re-engineering the AI content into human-centric, algorithm-friendly formats**.

1. Optimize Your NotebookLM Input & Prompting

NotebookLM's video overviews (like Video Overviews or Cinematic Video Overviews) are only as good as the raw sources you feed them.

- **Curate High-Quality Sources:** Feed the notebook full research papers, verified tech specs, or expert articles rather than brief summaries. The deeper the source data, the higher the accuracy.
- **Guide the Output Generation:** Don't just click generate. Use directional prompts in NotebookLM, such as:

"Create a 45-second script explaining this scientific discovery to a non-technical audience. Start with a shocking visual hook, simplify the core mechanism, and end with how it changes the future."
- **Fact-Check AI Output:** Science/Tech audiences on Facebook are hyper-critical. Always double-check citations and facts in NotebookLM before rendering your final export, as a single hallucinated detail will destroy credibility in your comment section.

2. Tailor Video Formatting for Facebook Reels

Raw AI exports rarely fit Facebook's ideal Reel specifications out of the box. You must adjust them in a secondary video editor (like CapCut or Premiere):

- **Enforce a Strict 9:16 Aspect Ratio:** Ensure the generated video completely fills vertical mobile screens (1080 x 1920 px).
- **Add Human Hook Overlays:** Take the first 3 seconds generated by NotebookLM and overlay a bold text title (e.g., *"Scientists Just Broke Physics?"* or *"This Tech Will Replace Smartphones"*).
- **Speed Up the Pacing:** AI audio narration can sound slightly conversational or slow. Boost audio speed by 1.1x–1.15x and trim long pauses to maintain a high retention rate.
- **Keep Length Between 20 to 45 Seconds:** In the Science & Tech niche, concise, single-concept Reels outperform long, multi-point lectures.

3. Overcome Meta's "Unoriginal Content" Filters

Meta actively downranks purely automated, unedited AI content. To ensure your page gets organic recommendation reach:

- **Add Transformative Value:** Edit the AI video by adding custom text overlays, native Facebook sound effects, visual zoom cuts, or a personal introduction/voiceover layer.
- **Avoid Raw AI Watermarks:** If your NotebookLM output contains watermarks or platform logos, crop or edit them out. Meta penalizes branded watermarks.

4. Leverage Social Search & Community Engagement

Science and technology are naturally high-debate topics. Use this to trigger Meta's "Meaningful Engagement" signal:

- **Trigger the Comment Section:** End your videos with a discussion question rather than a simple call-to-action:
 - *Bad:* "Follow for more tech news!"
 - *Good:* "Would you let an AI perform surgery on you? Let me know below."
- **SEO-Optimize Your Captions:** Write 2–3 sentences in the caption using natural keywords (e.g., *Quantum Computing, Artificial Intelligence, Space Exploration, Meta Tech News*). Facebook uses visual-text modeling to categorize and serve your Reels to users interested in those specific topics.
- **Post Consistently:** Aim for 1 Reel per day (or a minimum of 4–5 per week). The testing algorithm needs consistent data points to identify which specific audience sub-niche (e.g., AI enthusiasts vs. astronomy fans) responds best to your page.

Sources

1. [NotebookLM Cinematic Video Overview: Full Guide \(2026\)](#)
2. [How to Use NotebookLM in 2026: A Practical Guide to Google's](#)

User prompt: Outline a step-by-step daily content pipeline from gathering science research to publishing the final Reel.

Response: To consistently publish high-performing Science & Tech Reels without spending hours editing every day, you need a repeatable system. This 6-step daily pipeline takes you from raw research to a fully optimized, published Facebook Reel in under 45 minutes.

1 Research & Source Gathering

Time: 10 mins | Tool: RSS / Science Portals

Collect 1–2 high-quality sources on a single compelling topic. Avoid broad overviews—focus on a specific discovery, breakthrough, or debate.

- **Where to source:** ArXiv, Nature News, ScienceDaily, MIT Technology Review, or TechCrunch.
- **Rule:** Download the PDF or save full article URLs. The deeper the original context, the better NotebookLM performs.

2 NotebookLM Processing & Script Generation

Time: 5 mins | Tool: NotebookLM

Upload your source docs into a fresh NotebookLM notebook. Run a focused prompt to generate a concise, viral-ready script structure.

PROMPT TO USE:

"Act as a science communicator. Using only the attached sources, create a 30-second script for a video. Structure:

1. Hook: A controversial or high-curiosity statement (1 sentence).
2. Mechanism: How it works in plain terms (2-3 sentences).
3. Impact: Why this matters for the future (1 sentence).
4. Debate Question: Ask the audience for their opinion to drive comments."

3 Audio & Video Asset Generation

Time: 10 mins | Tool: NotebookLM / AI Video Engine

Generate the visual and audio assets using NotebookLM's overview/video generation feature or export the script to your preferred voiceover generator.

- **Audio check:** Ensure the voice track is crisp, energetic, and free of mispronounced technical jargon (e.g., "LiDAR," "CRISPR").
- **Visual check:** Verify the raw generated clip illustrates the core subject (e.g., space hardware, neural networks, lab settings).

4 Human Editing & Formatting

Time: 10 mins | Tool: CapCut or Premiere (Mobile/Desktop)

Import the assets into your video editor to strip away "AI stiffness" and adapt it strictly for Facebook's Reels format.

- **Aspect Ratio:** Format to **9:16 vertical (1080 x 1920 px)**.
- **Pacing:** Increase audio speed to **1.1x** and cut out silent pauses between sentences.

- **Text Overlays:** Add dynamic, auto-generated captions in the upper-middle frame. Highlight key tech terms in yellow or green.
- **Pattern Interrupts:** Insert a visual cut, zoom, or b-roll overlay every 2–3 seconds.

5 Metadata & SEO Captioning

Time: 5 mins | Tool: Meta Business Suite

Prepare the upload on Meta Business Suite to schedule or publish directly.

- **Hook Caption:** Start the caption with a strong statement matching your video hook.
- **SEO Keywords:** Include 3–5 relevant keywords naturally in the text (e.g., *#ArtificialIntelligence, Quantum Computing, NASA, Tech News*).
- **Audio:** Select a rising/trending audio track from Facebook's library at low volume (5-10%) under your main voiceover to trigger trending audio signals.

6 Publish & Community Trigger

Time: 5 mins | Tool: Facebook Mobile App

Publish during peak active hours for your target demographic (typically 12:00 PM – 2:00 PM or 6:00 PM – 9:00 PM local time).

- **First Comment:** As soon as it goes live, post a pinned comment expanding on an extra fun fact or linking your primary source.
- **Early Engagement:** Respond to the first 5–10 comments within 30 minutes to signal active discussion to the algorithm.

Daily Pipeline Time Breakdown

Phase	Task	Duration
Input	Sourcing & NotebookLM Scripting	15 mins
Production	Asset Generation & Editing	20 mins
Distribution	Publishing, SEO & Comment Seed	10 mins
Total Daily Investment		45 mins

User prompt: Give me a set of specialized NotebookLM prompt templates designed for science and technology research papers.

Response: Here are four specialized prompt templates designed specifically for science and technology research papers. You can paste these directly into NotebookLM's chat or source query box after uploading your research PDFs or web links.

Template 1: The "Breakthrough & Debunking" Prompt

Best for: Controversial news, trending breakthroughs, or click-worthy tech updates.

Act as a world-class science communicator. Analyze the uploaded research paper and extract the single most surprising or

Draft a 30-to-45-second Facebook Reel script following this structure:

1. HOOK (0-3s): A bold, 1-sentence statement that challenges a common belief or reveals a shocking stat (e.g., "Everythi
2. THE BREAKTHROUGH (3-15s): Explain what the researchers actually did or discovered in extremely plain, non-academic te
3. THE MECHANISM (15-25s): How does it work? Use ONE clear real-world analogy.
4. FUTURE IMPACT (25-35s): How will this change daily life, medicine, or tech in the next 5-10 years?
5. ENGAGEMENT QUESTION (35-40s): A debate-provoking question to ask in the comment section.

Important: Avoid academic jargon (e.g., replace "heterogeneous cataloging" with "organizing different types of data").

Template 2: The "How It Works" Deep-Dive Prompt

Best for: AI tools, novel hardware, robotics, or engineering marvels.

Analyze the attached paper/doc about [Insert Tech Name, e.g., Solid-State Batteries / LLM Optimization].

Generate a fast-paced, educational Reel script structured as follows:

- TITLE (On-screen text hook): "How [Tech Name] Actually Works in 30 Seconds"
- SECTION 1: The Problem (0-5s): What is broken or slow with current technology?
- SECTION 2: The Innovation (5-18s): What did the authors invent/discover to fix it? Break it down into 2 clear steps.
- SECTION 3: The Catch (18-25s): What is the current limitation, cost, or bottleneck preventing it from being used every
- CALL TO ACTION (25-30s): "Save this video for when this technology officially launches!"

Include visual cues in brackets [e.g., Show diagram of battery cell / Zoom in on code] for where to add visual cuts.

Template 3: The "Future Timeline" Prompt

Best for: Space exploration, biotech, quantum computing, or long-term innovations.

Based on the attached research, construct a short timeline script forecasting how this discovery will unfold.

- Structure:
- HOOK: "Scientists just tested [Discovery], and here is what the next 10 years look like."
 - YEAR 1-2: What happens in the lab/testing phase right now?
 - YEAR 5: When will the first commercial prototypes or clinical trials start?
 - YEAR 10: How does this permanently alter society or industry once fully scaled?
 - OUTRO: "Would you trust/use this in 10 years? Comment below."

Keep sentences short (under 12 words per sentence) to ensure snappy audio delivery.

Template 4: The "David vs. Goliath / Industry Disruptor" Prompt

Best for: Open-source AI models, startup breakthroughs, or unexpected research findings.

Review the attached document and frame this research as a "Disruptor Story."

- Generate a script with this exact flow:
1. HOOK: "A team of researchers just achieved [Huge Milestone] for a fraction of the cost."
 2. THE CONTRAST: Compare what big industry leaders spent vs. what this paper's method accomplished.
 3. THE SECRET SAUCE: Explain the core trick or optimization technique they used in simple terms.
 4. WHY IT MATTERS: Explain why this levels the playing field for ordinary developers/scientists.
 5. CTA: "Drop 'PAPER' in the comments if you want the link to read the full study."

Pro-Tips for NotebookLM Science Scripts

- 1. **Request Analogies:** Science papers are dense. Always tell NotebookLM: *"Explain [Complex Term] using an analogy involving everyday objects."*
- 2. **Specify Tone:** Add *"Tone: Urgent, energetic, and curious"* to prevent the script from sounding like a university lecture.
- 3. **Word Count Limits:** Specify a max word count of **65–90 words total** for a 30-second Reel to account for natural speech pauses.

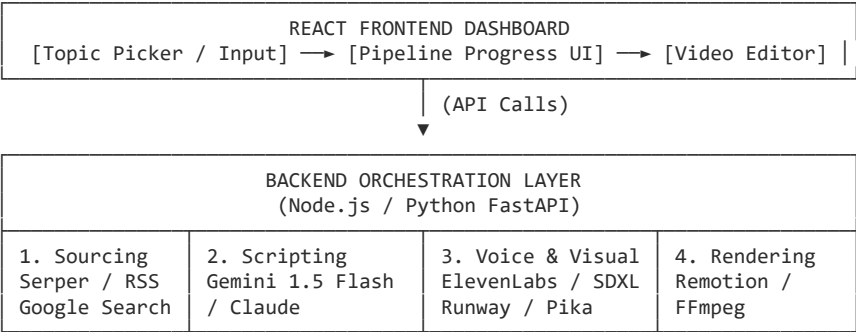
User prompt: i want to build react website to automate this process, i need follow things on this, pick trending topic or user will enter topic, and then go through the pipelinestep-by-step and generate the asset to build the reel video or generate the reel video, this is video content create platform could we do that?

Response: Yes, you can absolutely build this. What you are describing is an **Automated AI Video Creation SaaS Platform** tailored for short-form media (Reels/TikToks/Shorts).

Building this with **React** on the frontend paired with a backend orchestration service (like Node.js, Python/FastAPI, or Next.js serverless functions) is a well-established architecture for video automation.

Platform Architecture & Tech Stack

To handle video rendering and dynamic AI generation, your React app will act as the user dashboard while communicating with a backend pipeline through API services.



The 4-Step Technical Pipeline

Step 1: Topic Sourcing & Research

- **Trending Topics:** Use APIs like Google Trends API, Reddit API (/r/science, /r/technology), RSS feeds (Nature, MIT Tech Review), or Google News API via Serper/Brave Search API to fetch live trending headlines.

- **User Input:** Allow the user to paste a URL, research paper PDF, or custom prompt text.
- **AI Processing:** Pass the raw scraped text into **Gemini 1.5 Flash** (optimized for low latency and large context windows) to synthesize research points into a structured JSON payload.

Step 2: Scripting & Asset Structuring

- Pass the research summary to an LLM prompt structured to output strict JSON containing:
 - `hook_text` (for on-screen display)
 - `voiceover_script` (timestamped line by line)
 - `visual_prompts` (descriptions of B-roll needed for each timestamp)
 - `caption_keywords` (for SEO)

Step 3: Asset Generation

- **Voice Synthesis:** Send `voiceover_script` to **ElevenLabs API** or **OpenAI Audio API** to generate a natural .mp3 voice track.
- **B-roll / Visuals:**
 - *Option A (Cost-effective):* Query royalty-free stock APIs (**Pexels API**, **Pixabay API**) matching your visual keywords.
 - *Option B (AI Video/Image):* Pass `visual_prompts` to **Runway Gen-2 API**, **Luma Dream Machine API**, or **Stable Diffusion / Midjourney API** to generate high-tech AI imagery/video clips.
- **Subtitles:** Generate aligned .srt timestamped word files using **OpenAI Whisper API**.

Step 4: Video Assembly & Rendering (The Engine)

To stitch the audio, video clips, text overlays, and transitions into a 9:16 vertical video programmatically, you have two main options:

1. Remotion (Recommended for React developers):

- Remotion allows you to create videos programmatically using standard **React code** (HTML, CSS, Tailwind).
- You can define layout rules, animations, and lower-thirds in React, and render the final MP4 server-side (via Remotion Lambda) or in the browser.

2. FFmpeg Cloud Service:

- Use an API like **Shotstack API** or an backend **FFmpeg worker** on AWS EC2/Docker to programmatically overlay text and audio onto video clips.

Step-by-Step UI Workflow in React

[1. Topic Selection] → [2. Script Review & Edit] → [3. Asset Preview] → [4. Final Render]
 (Trends / URL / Text) (Editable Script JSON) (Audio/Clips Grid) (Export .MP4)

1. **Dashboard / Input:** Toggle between "Pick from Today's Tech Trends" or "Paste Paper / Custom Topic".
2. **Script Editor Component:** Displays the generated script in an editable layout so the user can tweak the hook, sentence pacing, or debate question before generating video assets.
3. **Timeline / Preview Stage:** Shows a preview player (using Remotion Player in React) where users can preview the voice track, swap out B-roll clips, or change text styles (font, colors, animation effects).
4. **Export & Direct Publish:** Render the final 1080x1920 MP4 file and allow direct downloading or publishing to Facebook via the **Meta Graph API**.

Key Challenges & Mitigation Strategies

- **Rendering Latency:** Video rendering takes compute power. Use a queue system (like BullMQ with Redis) on your backend to handle render jobs asynchronously so the React UI remains responsive.
- **API Cost Management:** Image and voice AI calls add up. Cache stock video clips by keyword and offer free stock assets with an option to upgrade to AI-generated visuals.
- **Audio-Visual Sync:** Ensure voiceover word-level timestamps (via Whisper) drive the timing of on-screen text overlays so text highlights perfectly in sync with speech.

User prompt: earlier you gav me different kind of specialized NotebookLM prompt templates designed for science and technology research papers, us those prompt techniques to more improvise this platform to give better result users to make better video, and use better optimized pipelines to workaround this, so give me full plan what we are going to do, whats on it, how we do it, i need to build mvp first

Response: To build this MVP efficiently, we need to transform the specialized narrative prompts into the core engine of your application. Instead of a generic "summarize this" tool, your platform will act as an "AI Video Director" that forces scientific data into high-retention Facebook Reel formats.

Here is the complete blueprint for building the MVP, focusing on low latency, cost-efficiency, and high-quality outputs.

Part 1: The Product Concept (What's in it?)

Your MVP will be a web-based dashboard where creators upload research and generate videos based on proven psychological hooks.

The Core UI Flow:

1. **The Ingestion Layer:** The user pastes a research URL, raw text, or uploads a document.
2. **The "Director" Selection:** The user selects a Narrative Framework (based on the specialized prompts):
 - *Mode 1: The Debunker* (Challenges a common belief)
 - *Mode 2: The How-It-Works* (Steps and mechanisms)
 - *Mode 3: The Future Timeline* (Forecasts and impact)
 - *Mode 4: The Disruptor Story* (David vs. Goliath tech)
3. **The Script Editor:** The app generates a structured, editable script broken into scenes.
4. **The Asset Board:** The app automatically pairs each scene with a voiceover and background video clips. The user can swap these out using client-side array filtering to manage and preview local/fetched media assets instantly.
5. **The Render Engine:** The user previews the 9:16 video in the browser and clicks "Export" to generate the final MP4.

Part 2: The Optimized Pipeline (How we work around bottlenecks)

Video generation platforms often fail because AI video models (like Runway or Sora) take minutes to generate a single clip, creating a terrible user experience. Here is the optimized workaround for the MVP.

Step 1: Low-Latency Text Processing

To eliminate long wait times, route the initial text analysis through high-speed models. **Gemini 1.5 Flash** (or Flash-Lite) is ideal here. It has a massive context window for reading full research papers instantly and is optimized for low-latency JSON extraction.

You will inject the user's selected Narrative Framework into the system prompt, forcing the LLM to output a strict JSON array.

Example LLM Output:

```
{
  "title": "Solid State Batteries Explained",
  "hook": "Everything we thought about electric cars just changed.",
  "scenes": [
    {
      "duration_seconds": 4,
      "spoken_text": "Everything we thought about electric cars just changed.",
      "broll_search_keyword": "electric car speeding futuristic",
      "on_screen_text": "EVs just changed forever."
    }
  ]
}
```

Step 2: The B-Roll Workaround (Speed > AI Generation)

Do not use AI to generate the video clips for the MVP. It is too slow and expensive.

- Instead, take the `broll_search_keyword` from the JSON and ping the **Pexels Video API** or **Pixabay API**.
- Fetch 3-4 royalty-free stock clips per scene instantly.
- This reduces scene generation time from 2 minutes (AI) to 400 milliseconds (API fetch).

Step 3: Audio & Timing Synchronization

- Send the `spoken_text` to a Text-to-Speech API (like ElevenLabs or OpenAI TTS).
- **The Optimization:** Request word-level timestamps from the audio API. This data tells your frontend exactly when to highlight words on the screen, creating dynamic, fast-paced captions that hold viewer attention.

Step 4: Client-Side Rendering with Remotion

Instead of queuing render jobs on expensive cloud servers, use **Remotion**. It allows you to build the video timeline using React components (HTML/CSS).

- The user previews the video perfectly in the browser.
- When they click "Export," you can render the video using an AWS Lambda function or a lightweight backend process, dropping the finalized MP4 into their dashboard.

Part 3: The MVP Tech Stack

To build this rapidly and robustly, structure your application using these technologies:

1. Backend & Database Layer

- **Framework:** C# with ASP.NET Core Web API. This provides a robust, strongly typed backend to handle API routing, secure API key management (for OpenAI/ElevenLabs), and user authentication.
- **Database:** SQL Server. Design tables to store user profiles, raw document text, generated JSON scripts, and project metadata.
- **Localization:** If you plan to support multiple languages for generated Reels, you can implement a database-backed localization utility to map translated UI elements or system prompts based on user preferences.

2. Frontend Application

- **Framework:** React.js.
- **Styling:** Custom CSS/HTML, utilizing responsive design principles (like Bootstrap or custom media queries) to ensure the video editor control panel scales cleanly from desktop down to mobile breakpoints (below 425px).
- **Video Engine:** Remotion (remotion.dev). This is the bridge that turns your React state into a playable, exportable video timeline.

3. API Integrations

- **Intelligence:** Gemini 1.5 Flash (for high-speed script generation and structuring).
- **Voice:** ElevenLabs (for hyper-realistic AI voiceovers).
- **Visuals:** Pexels Video API (for instant, high-quality B-roll).

Phase 1 Execution Plan (Next Steps)

1. **Database & API Setup:** Scaffold the SQL Server database with a `Projects` table and a `Scripts` table. Build the C# API endpoints to accept the initial text/URL payload.
2. **Prompt Engineering the Engine:** Hardcode the 4 Narrative Templates into your backend services. Ensure they reliably return the exact JSON structure needed for the frontend.
3. **The React Editor:** Build the UI where the user can view the JSON script blocks, click to regenerate specific lines, and filter through the fetched B-roll options.
4. **The Remotion Player:** Map the script JSON to a Remotion `<Composition>` component. Add a 9:16 layout constraint and sync the audio timestamps to the text overlays.

User prompt: user will only have to select things rather than typing or entering, we have to suggest things to select, we always giving best and very well optimize things to select, most of the time we suggest our order then user will alter if needed, this is how it should arranged, need to do this with minimum interaction from the user, but everything should be transparent and reasoning for each selection and ordering from through the platform, then user know what we are doing

Response: This "zero-typing, high-transparency" approach shifts the platform from a basic utility to an intelligent, opinionated "Auto-Director." By presenting pre-optimized choices and clearly explaining the *why* behind them, you eliminate friction while building user trust.

To achieve this, the C# backend and Gemini 1.5 Flash engine must be configured to generate not just the content, but the **metadata and reasoning** for every decision.

Here is how to structure this automated, highly transparent MVP pipeline.

The "Zero-Typing" Pipeline Architecture

Stage 1: The Daily Topic Feed (Zero-Input Sourcing)

Instead of an empty input box, the user logs into a dashboard populated with today's top science and tech topics.

- **How it works:** A background C# service runs periodically, scraping RSS feeds (e.g., Nature, TechCrunch) or the Google Trends API. It stores these in a SQL Server `TrendingTopics` table.
- **The User Experience:** The React UI displays a grid of Bootstrap cards containing the top 5 trending topics.

- **The Transparency Layer:** Each card includes a "Why this?" badge.
 - *Example:* "Ranked #1: 'Solid-State Battery Breakthrough' has seen a 400% search volume increase on Google Trends in the last 24 hours."

Stage 2: The Auto-Director (Framework Matching)

When the user clicks a topic card, the system automatically assigns the best Narrative Framework (Debunker, How-It-Works, etc.) instead of making the user guess.

- **How it works:** You send the source article text to Gemini 1.5 Flash with a classification prompt. The prompt forces the LLM to output a JSON object containing the selected framework and a `reasoning_string`.
- **The User Experience:** The UI automatically transitions to the next step, showing the selected framework at the top.
- **The Transparency Layer:** A small info alert explains the selection.
 - *Example:* "We selected **The Disruptor Story** template because the source article compares a small startup's new microchip directly against Nvidia's standard."

Stage 3: The Auto-Populated Storyboard (Review & Alter)

This is the core control panel. The system generates the entire script, splits it into scenes, and pre-selects the background B-roll and audio pacing.

- **How it works:** The backend calls Gemini to generate the scene-by-scene JSON. It immediately uses the extracted keywords to query the Pexels API, pulling 5 video options per scene. It pre-selects the first one.
- **The User Experience:** A responsive storyboard layout. On desktop, this is a multi-column grid; on mobile screens (scaling cleanly down to 425px widths), it stacks vertically. Each scene shows the script text, the assigned voiceover length, and the selected B-roll thumbnail.
- **The Transparency Layer:** Each scene block explains its technical optimizations.
 - *Example:* "Audio sped up to 1.1x to keep this scene under 4 seconds (the optimal retention threshold)."
- **The "Alteration" Mechanism:** If the user dislikes the auto-selected B-roll, they click a "Swap Visual" button. Because you already fetched 5 options from the API, you can use client-side JavaScript array filtering to instantly display alternative thumbnails without needing another server round-trip.

Stage 4: One-Click Render

Because all selections were pre-optimized and user-approved in Stage 3, the final step is simply hitting "Render." The backend orchestrates the Remotion build and outputs the MP4.

Engineering the "Transparency JSON"

The key to making this work is strictly formatting your prompts to Gemini 1.5 Flash. You must force the LLM to provide its internal logic alongside the content.

Here is the exact JSON schema your C# API should request from Gemini:

```
{
  "selected_framework": "how_it_works",
  "framework_reasoning": "The source document details a step-by-step mechanical process, making the 'How It Works' struc",
  "suggested_title": "How Quantum Sensors Actually Work",
  "scenes": [
    {
      "scene_id": 1,
      "hook_text": "Everything you know about GPS is about to change.",
      "script_reasoning": "Uses a 'Negative Warning' hook to immediately interrupt scrolling.",
      "broll_search_term": "satellite orbit earth",
      "visual_reasoning": "A macro shot of a satellite establishes high-tech context within the first 1.5 seconds."
    }
  ]
}
```

Database Schema Adjustments (SQL Server)

To support this transparent, pre-calculated workflow, your database structure will need specific tables:

1. **RawSources:** Stores the scraped RSS/Trends data (URL, Headline, Date, Search Volume) so the Daily Feed loads instantly.
2. **GeneratedProjects:** Stores the user's active video project.
3. **ProjectScenes:** Relational table linking back to the project. Stores the `hook_text`, `broll_search_term`, and crucially, the `script_reasoning` text so the React UI can display the transparency tooltips directly from the database query.

By treating the AI's *reasoning* as a first-class data point stored in your database and displayed in the UI, you transform a potentially frustrating "black box" AI tool into an expert assistant that guides the user to the best possible result.

User prompt: i should select react, node and mysql for frontend, backend and database, first of all lets create a detail report about what we discussed here, idea, and include what we are going to do, how we going to it, dont miss any single small detail

Response: System Architecture & Product Blueprint: AI-Automated Video Director SaaS

1. Executive Summary

The platform is an "Auto-Director" SaaS application designed to automate the creation of highly optimized, short-form Science & Technology video content for Facebook Reels.

Unlike standard "text-to-video" generators that require heavy user prompting and produce slow, unpredictable results, this platform uses a **"zero-typing, high-transparency" workflow**. The system ingests trending tech news, autonomously selects the optimal viral narrative structure, generates the script, and pre-assembles the video assets. The user simply reviews the AI's logic, makes one-click alterations, and renders.

2. Technical Stack Definition

The infrastructure transitions from the previously discussed C#/SQL Server stack to a modern JavaScript ecosystem to unify the codebase language across the video renderer, frontend, and backend.

- **Frontend:** React.js (utilizing Bootstrap for UI components and custom media queries for responsive layouts down to mobile 425px widths).
- **Backend Orchestration:** Node.js (with Express.js or Fastify) for handling API routing, background cron jobs, and secure third-party API key management.
- **Database:** MySQL (Relational structure for strict project and asset mapping).
- **Video Rendering Engine:** Remotion (React-based programmatic video generation).
- **AI & APIs:**
 - *Logic & Scripting:* Gemini 1.5 Flash (optimized for low latency, large context ingestion, and strict JSON output).
 - *Voice Generation:* ElevenLabs or OpenAI TTS (must support word-level timestamps).
 - *Visual Assets:* Pexels Video API or Pixabay API (for instantaneous B-roll fetching).

3. The Meta/Facebook Optimization Ruleset

All platform outputs are hardcoded to satisfy the Facebook Reels algorithm. The final MP4 outputs will adhere to:

- **Format:** Strictly 9:16 vertical ratio (1080 x 1920 px).
- **Pacing:** Spoken audio sped up to 1.1x–1.15x. Visual pattern interrupts (cuts/zooms) every 2–3 seconds.
- **The Hook:** High-impact text placed in the screen's center-upper third (the "safe zone"), paired with immediate visual movement.
- **Algorithmic Signals:**
 - On-screen text synced perfectly with audio to retain muted viewers.
 - Call-to-actions designed to trigger comments/debate, not just likes.
 - Captions formatted with natural language SEO keywords for Meta's visual-text index.

4. Core Application Workflow (Zero-Typing & High Transparency)

The user journey is designed to minimize input fatigue while building trust through explanatory tooltips.

Stage 1: The Daily Topic Feed

- **Backend:** A Node.js cron job scrapes RSS feeds (Nature, MIT Tech Review) and the Google Trends API, saving trending topics to the MySQL database.
- **Frontend:** Users log in to a dashboard of Bootstrap cards displaying today's topics.
- **Transparency:** Each card includes a badge explaining *why* it was pulled (e.g., "Trending: 400% search volume spike in 24 hours").

Stage 2: The Auto-Director (Framework Selection)

- **Backend:** Upon selecting a topic, Node.js sends the source material to Gemini 1.5 Flash. Gemini categorizes the topic into one of four hardcoded frameworks: *The Debunker*, *The How-It-Works*, *The Future Timeline*, or *The Disruptor Story*.
- **Transparency:** The UI transitions and displays an alert: "Selected *The Disruptor Story* framework because the source compares a startup directly against industry giants."

Stage 3: Auto-Populated Storyboard

- **Backend:** Gemini generates a highly structured JSON object containing the scene-by-scene script, hooks, and B-roll search terms. Node.js immediately queries the Pexels API using those terms, fetching an array of 5 video options per scene.
- **Frontend:** The React UI renders a vertical or grid-based storyboard. Each scene block contains the spoken text, estimated duration, and the pre-selected B-roll thumbnail.
- **Transparency:** UI tooltips explain scene-level decisions (e.g., "Audio sped up to keep this hook under 3 seconds").
- **Alteration:** If the user dislikes the visual, they click "Swap". Client-side JavaScript array filtering instantly cycles through the other 4 fetched Pexels clips without triggering a new backend API call.

Stage 4: Sync & Render

- **Backend:** The selected script is sent to the TTS API. The API returns the .mp3 and an array of word-level timestamps.
- **Frontend:** Remotion mounts the video in the browser. The word-level timestamps drive the React text-highlighting components so the on-screen captions match the audio perfectly.
- **Action:** The user previews the video and clicks "Render," generating the final MP4 either via a background Node.js worker or directly in the browser (depending on Remotion configuration).

5. MySQL Database Schema (Initial Blueprint)

The relational database will connect the scraped data, user projects, and the AI's internal reasoning.

Table Name	Primary Purpose	Key Columns
TrendingSources	Stores scraped RSS/News articles for the daily feed.	SourceID, Headline, URL, RawText, TrendReasoning, CreatedAt
Projects	Tracks the user's active video generation sessions.	ProjectID, UserID, SourceID, SelectedFramework, FrameworkReasoning, Status
ProjectScenes	The granular scene-by-scene script and asset data.	SceneID, ProjectID, SceneOrder, SpokenText, ScriptReasoning, BRollSearchTerm
SceneAssets	Caches the Pexels API URLs to avoid re-fetching.	AssetID, SceneID, VideoURL, ThumbnailURL, IsSelected

6. The Gemini JSON Engine

The Node.js backend will enforce this exact JSON schema when communicating with Gemini 1.5 Flash. This schema guarantees both the content and the transparency reasoning are delivered in one payload:

```
{
  "selected_framework": "how_it_works",
  "framework_reasoning": "The source details a mechanical process.",
  "suggested_title": "Quantum Sensors Explained",
  "seo_caption": "Quantum sensors are changing navigation... #Quantum #TechNews",
  "scenes": [
    {
      "scene_order": 1,
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      "script_reasoning": "Uses a 'Negative Warning' hook.",
      "broll_search_term": "satellite orbit earth",
      "visual_reasoning": "Establishes high-tech context immediately."
    }
  ]
}
```

User prompt: give me md file including all these details

Response: Here is the complete Markdown file containing the system architecture, product blueprint, and technical specifications. You can copy this code block and save it directly as a .md file (e.g., video-director-blueprint.md).

System Architecture & Product Blueprint: AI-Automated Video Director SaaS

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- * **Video Rendering Engine:** Remotion (React-based programmatic video generation for in-browser preview and serverles
- * **AI & APIs:**
 - * **Logic & Scripting:** Gemini 1.5 Flash (or flash-lite-preview versions) to strictly prioritize processing speed a
 - * **Voice Generation:** ElevenLabs or OpenAI TTS (must support word-level timestamps).
 - * **Visual Assets:** Pexels Video API or Pixabay API (for instantaneous B-roll fetching).

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Stage 3: Auto-Populated Storyboard & Asset Filtering

- * **Backend:** Gemini generates a highly structured JSON object containing the scene-by-scene script, hooks, and B-rol
- * **Frontend:** The React UI renders a responsive storyboard. Each scene block contains the spoken text, estimated dur
- * **Custom Capture:** The UI includes an HTML file input configured with the `capture="environment"` attribute, allowi
- * **Transparency Layer:** UI tooltips explain scene-level decisions (e.g., "Audio sped up to keep this hook under 3 se
- * **Instant Alteration:** If the user dislikes the visual, they click "Swap". Client-side JavaScript array filtering i

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- * **Frontend:** Remotion mounts the video in the browser. The word-level timestamps drive the React text-highlighting
- * **Action:** The user previews the video and clicks "Render," generating the final MP4.

5. MySQL Database Schema

The relational database connects scraped data, user projects, AI logic, and includes foundational tables for a database-

```
```sql
-- Topic Sourcing
CREATE TABLE TrendingSources (
 SourceID INT AUTO_INCREMENT PRIMARY KEY,
 Headline VARCHAR(255),
 SourceURL TEXT,
 RawContent TEXT,
 TrendReasoning VARCHAR(255),
 CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```

-- User Video Projects
CREATE TABLE Projects (
 ProjectID INT AUTO_INCREMENT PRIMARY KEY,
 UserID INT,
 SourceID INT,
 SelectedFramework VARCHAR(50),
 FrameworkReasoning TEXT,
 TargetLanguage VARCHAR(10) DEFAULT 'en',
 Status VARCHAR(20) DEFAULT 'Draft',
 FOREIGN KEY (SourceID) REFERENCES TrendingSources(SourceID)
);

-- AI Generated Script & Transparency Data
CREATE TABLE ProjectScenes (
 SceneID INT AUTO_INCREMENT PRIMARY KEY,
 ProjectID INT,
 SceneOrder INT,
 SpokenText TEXT,
 ScriptReasoning TEXT,
 BRollSearchTerm VARCHAR(100),
 VisualReasoning TEXT,
 FOREIGN KEY (ProjectID) REFERENCES Projects(ProjectID)
);

-- Cached B-Roll Assets for Client-Side Filtering
CREATE TABLE SceneAssets (
 AssetID INT AUTO_INCREMENT PRIMARY KEY,
 SceneID INT,
 VideoURL TEXT,
 ThumbnailURL TEXT,
 IsSelected BOOLEAN DEFAULT FALSE,
 FOREIGN KEY (SceneID) REFERENCES ProjectScenes(SceneID)
);

-- Localization / Translation Dictionary
CREATE TABLE SystemTranslations (
 TranslationKey VARCHAR(100),
 LanguageCode VARCHAR(10),
 TranslatedText TEXT,
 PRIMARY KEY (TranslationKey, LanguageCode)
);

```

## 6. The AI JSON Engine Payload

The Node.js backend enforces this exact schema when prompting Gemini 1.5 Flash. This guarantees the content and the transparency reasoning are delivered in one optimized payload:

```

{
 "selected_framework": "how_it_works",
 "framework_reasoning": "The source document details a step-by-step mechanical process, making the 'How It Works' struc",
 "suggested_title": "How Quantum Sensors Actually Work",
 "seo_caption": "Quantum sensors are changing navigation... #Quantum #TechNews",
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 }
]
}

```

<FollowUp label="Want to draft the React Remotion component next?" query="Show me how to structure the React Remotion co

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